Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030001

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Query 1: Create the database

```
CREATE DATABASE IF NOT EXISTS sql_db;
USE sql_db;
```

Output: -

3	09:44:54	CREATE DATABASE IF NOT EXISTS sql_db	1 row(s) affected	0.000 sec
4	09:44:58	USE sql_db	0 row(s) affected	0.016 sec

Query 2: -CREATE PHYSICIAN TABLE

```
-- 2520030001-Aswitha
CREATE TABLE physician (
    employeeid INT PRIMARY KEY,
    name VARCHAR(100),
    position VARCHAR(100),
    ssn VARCHAR(20)
);
```

Output: -

5	09:46:21	CREATE TABLE physician (employeeid INT PRI...	0 row(s) affected	0.063 sec

Query 3: -INSERT DATA INTO PHYSICIAN

```sql
-- 2520030001-Aswitha
INSERT INTO physician VALUES
(1,'M.Aswitha Patel','Staff Internist','111111111'),
(2,'M.Adhritha','Attending Physician','222222222'),
(3,'Akshara Reddy','Surgical Attending Physician','333333333'),
(4,'Anirudh Reddy','Senior Attending Physician','444444444'),
(5,'Adithya Vardhan','Head Chief of Medicine','555555555'),
(6,'Ananya Varshitha','Surgical Attending Physician','666666666
(7,'Yashwanth Reddy','Surgical Attending Physician','777777777'
(8,'Nihitha','MD Resident','888888888'),
(9,'Ramya Sree','Attending Psychiatrist','999999999');
```

Output: -

9	14:32:03	INSERT INTO physician VALUES (1,'M.Aswitha P...	9 row(s) affected Records: 9 Duplicates: 0 Warnin...	0.016 sec

Query 4: -CREATE DEPARTMENT TABLE

```sql
CREATE TABLE department (
    departmentid INT PRIMARY KEY,
    name VARCHAR(100),
    head INT
);
```

Output: -

10	14:33:35	CREATE TABLE department (departmentid INT ...	0 row(s) affected	0.047 sec

Query 5: -INSERT DATA INTO DEPARTMENT

```sql
INSERT INTO department VALUES
(1,'General Medicine',4),
(2,'Surgery',7),
(3,'Psychiatry',9);
```

Output: -

11	14:35:14	INSERT INTO department VALUES (1,'General Me...	3 row(s) affected Records: 3 Duplicates: 0 Warnin...	0.016 sec

Query 6: -INNER JOIN

```
-- 2520030001-Aswitha
SELECT d.name AS Department,
       p.name AS Head_Physician
FROM department d
JOIN physician p
ON d.head = p.employeeid;
```

Output: -

12 14:36:41 SELECT d.name AS Department, p.name AS ... 3 row(s) returned 0.000 sec / 0.000 sec

Result Grid | Filter Rows:

Department	Head_Physician
General Medicine	Anirudh Reddy
Surgery	Yashwanth Reddy
Psychiatry	Ramya Sree

Query 7: - WHERE CLAUSE

```
-- 2520030001-Aswitha
SELECT *
FROM physician
WHERE position='Surgical Attending Physician';
```

Output: -

13 14:38:23 SELECT * FROM physician WHERE position='Sur... 3 row(s) returned 0.016 sec / 0.000 sec

employeeid	name	position	ssn
3	Akshara Reddy	Surgical Attending Physician	333333333
6	Ananya Varshitha	Surgical Attending Physician	666666666
7	Yashwanth Reddy	Surgical Attending Physician	777777777
NULL	NULL	NULL	NULL

Query 8: - WHERE + LIKE

```sql
-- 2520030001-Aswitha
SELECT *
FROM physician
WHERE name LIKE 'Nihitha%';
```

Output: -

15	14:40:11	SELECT * FROM physician WHERE name LIKE '...	1 row(s) returned	0.000 sec / 0.000 sec

employeeid	name	position	ssn
8	Nihitha	MD Resident	888888888
NULL	NULL	NULL	NULL

Query 9: - COUNT()

```sql
-- 2520030001-Aswitha
SELECT COUNT(*) AS Total_Physicians
FROM physician;
```

Output: -

16	14:42:16	SELECT COUNT(*) AS Total_Physicians FROM p...	1 row(s) returned	0.032 sec / 0.000 sec

Result Grid | Filter Row

Total_Physicians
9

Query 10: - COUNT DISTINCT

```sql
-- 2520030001-Aswitha
SELECT COUNT(DISTINCT position)
FROM physician;
```

Output: -

17	14:44:11	SELECT COUNT(DISTINCT position) FROM phys...	1 row(s) returned	0.000 sec / 0.000 sec

Result Grid | Filter Rows:

	COUNT(DISTINCT position)
▶	7

Query 11: -GROUP BY

```
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position;
```

Output: -

18 14:47:37 SELECT position, COUNT(*) AS Total_Emplo... 7 row(s) returned 0.016 sec / 0.000 sec

	position	Total_Employees
▶	Staff Internist	1
	Attending Physician	1
	Surgical Attending Physician	3
	Senior Attending Physician	1
	Head Chief of Medicine	1
	MD Resident	1
	Attending Psychiatrist	1

Query 12: -GROUP BY + HAVING

```
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
HAVING COUNT(*) > 1;
```

Output: -

19 14:49:06 SELECT position, COUNT(*) AS Total_Emplo... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid | Filter Rows: | Export:

	position	Total_Employees
▶	Surgical Attending Physician	3

Query 13: -ORDER BY

```
SELECT *
FROM physician
ORDER BY name ASC;
```

Output: -

20 14:50:34 SELECT * FROM physician ORDER BY name AS... 9 row(s) returned 0.032 sec / 0.000 sec

Result Grid | Filter Rows: | Edit: | Export/Im

	employeeid	name	position	ssn
▶	5	Adithya Vardhan	Head Chief of Medicine	555555555
	3	Akshara Reddy	Surgical Attending Physician	333333333
	6	Ananya Varshitha	Surgical Attending Physician	666666666
	4	Anirudh Reddy	Senior Attending Physician	444444444
	2	M.Adhritha	Attending Physician	222222222
	1	M.Aswitha Patel	Staff Internist	111111111
	8	Nihitha	MD Resident	888888888
	9	Ramya Sree	Attending Psychiatrist	999999999
	7	Yashwanth Reddy	Surgical Attending Physician	777777777
*	NULL	NULL	NULL	NULL

Query 14: -ORDER BY DESC

```
-- 2520030001-Aswitha
SELECT *
FROM physician
ORDER BY employeeid DESC;
```

Output: -

20 14:50:34 SELECT * FROM physician ORDER BY name AS... 9 row(s) returned 0.032 sec / 0.000 sec

employeeid	name	position	ssn
5	Adithya Vardhan	Head Chief of Medicine	555555555
3	Akshara Reddy	Surgical Attending Physician	333333333
6	Ananya Varshitha	Surgical Attending Physician	666666666
4	Anirudh Reddy	Senior Attending Physician	444444444
2	M.Adhritha	Attending Physician	222222222
1	M.Aswitha Patel	Staff Internist	111111111
8	Nihitha	MD Resident	888888888
9	Ramya Sree	Attending Psychiatrist	999999999
7	Yashwanth Reddy	Surgical Attending Physician	777777777
NULL	NULL	NULL	NULL

Query 15: - WHERE + IN

```
SELECT *
FROM physician
WHERE employeeid IN
(
    SELECT head
    FROM department
);
```

Output: -

21 14:54:58 SELECT * FROM physician WHERE employeeid I... 3 row(s) returned 0.000 sec / 0.000 sec

Result Grid Filter Rows: Edit: Export/Im

employeeid	name	position	ssn
4	Anirudh Reddy	Senior Attending Physician	444444444
7	Yashwanth Reddy	Surgical Attending Physician	777777777
9	Ramya Sree	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

Query 16: -SUBQUERY

```sql
-- 2520030001-Aswitha
SELECT *
FROM physician
WHERE employeeid NOT IN
(
    SELECT head
    FROM department
);
```

Output: -

22 14:56:50 SELECT * FROM physician WHERE employeeid ... 6 row(s) returned 0.032 sec / 0.000 sec

employeeid	name	position	ssn
1	M.Aswitha Patel	Staff Internist	111111111
2	M.Adhritha	Attending Physician	222222222
3	Akshara Reddy	Surgical Attending Physician	333333333
5	Adithya Vardhan	Head Chief of Medicine	555555555
6	Ananya Varshitha	Surgical Attending Physician	666666666
8	Nihitha	MD Resident	888888888
NULL	NULL	NULL	NULL

Query 17: - GROUP BY + ORDER BY

```sql
-- 25200300001-Aswitha
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
ORDER BY Total_Employees DESC;
```

Output: -

23 14:58:30 SELECT position, COUNT(*) AS Total_Emplo... 7 row(s) returned 0.000 sec / 0.000 sec

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Query 18: - JOIN + WHERE

```
--  2520030001-Aswitha
SELECT p.name
FROM physician p
JOIN department d
ON p.employeeid=d.head
WHERE d.name='Surgery';
```

Output: -

24 15:01:13 SELECT p.name FROM physician p JOIN depart... 1 row(s) returned 0.000 sec / 0.000 sec

name
Yashwanth Reddy

Query 19: - JOIN + ORDER BY

```
-- 2520030001-Aswitha
SELECT d.name,
       p.name
FROM department d
JOIN physician p
ON d.head=p.employeeid
ORDER BY d.name;
```

Output: -

25 15:02:46 SELECT d.name, p.name FROM department ... 3 row(s) returned 0.000 sec / 0.000 sec

name	name
General Medicine	Anirudh Reddy
Psychiatry	Ramya Sree
Surgery	Yashwanth Reddy

Query 20: - WHERE + NOT LIKE

```
SELECT *
FROM physician
WHERE position NOT LIKE '%Surgical%';
```

Output: -

26 15:04:17 SELECT * FROM physician WHERE position NO... 6 row(s) returned 0.000 sec / 0.000 se

employeeid	name	position	ssn
1	M.Aswitha Patel	Staff Internist	111111111
2	M.Adhritha	Attending Physician	222222222
4	Anirudh Reddy	Senior Attending Physician	444444444
5	Adithya Vardhan	Head Chief of Medicine	555555555
8	Nihitha	MD Resident	888888888
9	Ramya Sree	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

Query 21: - COUNT + HAVING

```
-- 2520030001-Aswitha
SELECT position,
       COUNT(*) AS Total
FROM physician
GROUP BY position
HAVING COUNT(*) >= 2;
```

Output: -

27 15:05:45 SELECT position, COUNT(*) AS Total FROM ... 1 row(s) returned 0.000 sec / 0.000 sec

position	Total
Surgical Attending Physician	3

Query 22: - EXISTS

```
HAVING COUNT(*) >= 2;
SELECT *
FROM department d
WHERE EXISTS
(
    SELECT 1
    FROM physician p
    WHERE p.employeeid=d.head
);
```

Output: -

28	15:07:03	SELECT * FROM department d WHERE EXISTS ...	3 row(s) returned	0.000 sec / 0.000 sec

Result Grid | Filter Rows:

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9
NULL	NULL	NULL

Query 23: - MAX()

```
-- 2520030001-Aswitha
SELECT MAX(employeeid)
FROM physician;
```

Output: -

29	15:08:41	SELECT MAX(employeeid) FROM physician LIMI...	1 row(s) returned	0.000 sec / 0.000 sec

MAX(employeeid)
9

Query 24: - MIN()

```
-- 2520030001-Aswitha
SELECT MIN(employeeid)
FROM physician;
```

Output: -

30	15:10:24	SELECT MIN(employeeid) FROM physician LIMIT...	1 row(s) returned	0.000 sec / 0.000 sec

Result Grid

	MIN(employeeid)
▶	1